Mathematical Modelling - NUIST

Engineering Technology

May, 2026

South East Technological University

Ollscoil Teicneolaíochta an Oirdheiscirt

Mathematical Modelling - NUIST (A35616)

Short Title:	Mathematical Modelling (NUIST)		
Faculty	Science and Computing		
Department	Engineering Technology		
Cluster	Science and Mathematics		
Author	CLAIRE.FITZPATRICK		
Credits	5	**Level:**	Postgraduate

Description of Module / Aims

This module is designed to take the student through the mathematical modelling process from specification to modelling and simulation. The focus is on stochastic modelling. Both discrete and continuous systems are treated including: queuing theory, additive white Gaussian noise (AWGN) in digital communications, discrete symbol alphabets. Analytic results for average behaviour are developed and these are investigated through numerical simulations in the mini-project. Associated mathematical ideas such as orthogonality are explored analytically through polynomials where recurrence relations, and weighting factors are involved.

Programmes

`MSTU-0001`	MEng in Electronic Engineering (`WD_EELEN_R`)	1 / 1 / M
	MEng in Electronic Information Systems (`WD_EELEN_RI`)	1 / 1 / M
`MSTU-0001`	PG Dip in Engineering in Electronic Engineering (`WD_EELEN_G`)	1 / 1 / M

Indicative Content

- Introduction to the modelling process: formulation, implementation, simulation, interpretation and evaluation of results.
- Stochastic processes and random variables: characterisation - probability density functions, autocorrelation, power spectral density and thermal noise: queuing theory
- Queuing systems: birth - death processes, M/M/1 queue, differential difference equations, pure birth - Poisson distribution
- Digital comms: DS-CDMA - effect of multiple users on BER performance, code length, COFDM: simulation in flat and fading channels

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Follow and reproduce simulations for selected problems in electronic engineering.
2. Implement numerical techniques to standard problems and interpret the results.
3. Apply theory to a problem, selecting and implementing an approach, interpreting and adapting to results.
4. Demonstrate modelling & simulation for assignments & mini project.

Learning and Teaching Methods

- Formal lectures will provide the theoretical base for the subject as well as providing and overview of issues that arise in application of theory. Case studies, with subsequent discussion and explanation, together with feedback from suggested reading will support and direct the students in their own personal study. Computer laboratory sessions will be used to demonstrate implementations of methods and provide framework for feedback and assistance on project work.
- The laboratory sessions consist of a number of computer based practicals to demonstrate the theory.

- The ability to apply the theory covered in this module to progress from a physical problem, to selecting and implementing an appropriate solution strategy, and ultimately interpreting the results is a fundamental learning objective of this module. The ability is assessed primarily through the practical work.

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	50%	1, 2, 3
Continuous Assessment	50%	
Project	40%	1, 2, 3, 4
In-Class Assessment	10%	1, 2, 3, 4

Assessment Criteria

70% - 100%: The learner has: attained the module learning outcomes at an excellent level; demonstrated a comprehensive knowledge of the associated subject; achieved an excellent level of the skills required for the subject matter; demonstrated the ability to carry out further investigation and problem solving associated with the subject matter.

60% - 69%: The learner has: attained the module learning outcomes at a very good level; demonstrated a detailed knowledge of the associated subject matter; achieved a very good level of the skills required for the subject matter.

40% - 59%: The learner has: attained the module learning outcomes at a good level; demonstrated a good knowledge of the associated subject matter; achieved a good level of the skills required for the subject matter.

<40%: The learner has not: attained the module learning outcomes; demonstrated sufficient knowledge of the associated subject matter; demonstrated a sufficient level of the skills required for the subject matter.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	24	
Practical	6	
Independent Learning	105	

Essential Material

- Connor, F.R. _Noise_. 2nd ed.. London: Edward Arnold, 1982.
- Kleinrock, L. _Queuing Systems_. New York: Wiley, 1975.

Supplementary Material

- Grimmett, G.R. and D.R. Stirzaker. _Probability and Random Processes_. 2nd ed.. Oxford: Oxford University Press, 1992.
- Primak, S. _Stochastic Methods and their Applications to Communications: Stochastic Differential Equations Approach_. New York: Wiley, 2004.

- Riley, K.F. _Mathematical Methods for Physics and Engineering: A__Compehensive Guide_. Cambridge: Cambridge University Press, 2002.
- Shannon, C. and W. Weaver. _A mathematical theory of communication_. New York: Univerisity of Illinois/Dover paperback, 1948.